NITHIN SKANTHA MURUGAN

AI Engineer & Founder · Autonomous Agents · Computer Vision · Real-time Systems

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01 EXPERIENCE

Renfort · Co-Founder & CEO — College Park, MD · **Feb 2026 – Present**

- Architecting autonomous AI agents that operate 20-year-old legacy dental EHRs, extracting structured fields from real-time ambient clinician–patient speech via direct schema writes and native Win32 UI automation.
- Engineered full-stack agent framework: .NET 8/WPF desktop shell, Fastify on Bun backend, Speechmatics real-time streaming, and multi-model LLM orchestration (GPT-5.5, Claude Sonnet).
- Deployed on live PHI across 7 dental practices (NV, TX, LA, MD) under HIPAA compliance, signed Renfort's **first paying customer**; secured **$35,000** in non-dilutive funding from MIT and University of Maryland.

Relai, Inc. · Field AI Engineer — Remote · **Jul 2026 – Sep 2026**

- Engineered continual learning pipelines for autonomous AI agents, improving model adaptability and memory retention across iterative deployments.
- Scoped, architected, and shipped enterprise proof-of-concept agent integrations for platforms like ServiceNow, automating complex workflows for high-volume enterprise clients.

DefenX · Co-Founder & Chief AI Officer — College Park, MD · **Sep 2024 – May 2026**

- Architected real-time AI safety infrastructure for K-12 schools, integrating weapon detection, multi-camera threat tracking, and emergency response coordination across active campus networks.
- Custom-trained YOLOv11 weapon detector with sub-1% false-positive rate and 6.7% accuracy gain over SOTA; designed 3D-calibrated multi-camera tracking across 25+ video feeds.
- Won the **$250,000 grand prize** at XFoundry Xperience; system deployed in active school environments.

University of Maryland · Graduate Researcher · Advisor: Prof. Tom Goldstein — College Park, MD · **Oct 2024 – Dec 2025**

- Led research team building cross-model likelihood frameworks for AI-generated content detection, reaching 92%+ accuracy across diffusion models, GANs, and autoregressive generators (LlamaGen).
- Benchmarked adversarial robustness of multimodal LLMs under physical and digital perturbations (JPEG compression, Gaussian noise) to design defensive evaluation metrics.

StumpEye Limited · Founding Computer Vision Engineer — Milton Keynes, UK (Remote) · **Jan 2023 – Jun 2024**

- Architected an end-to-end Decision Review System replacing $60K broadcast rigs with a $500 mobile system, using YOLOv8 ball detection (92%+ mAP) and Transformer trajectory prediction (MAE < 0.08).
- Engineered multi-threaded C++/OpenCV pipeline on ONNX/TensorFlow Lite for 30+ FPS real-time mobile inference; won the ICC Global Hackathon over 22,000+ entries from 119 countries.

02 PROJECTS

Cloud Orchestrator · Google ADK, Gemini 2.5, Flask, GCP — **Feb 2026**

- Built a multi-agent natural-language orchestrator generating execution DAGs for 20+ GCP services (Vertex AI, GKE, BigQuery), cutting cloud setup time by 95% (Google ADK Hackathon).

FeedForward · OpenAI Realtime API, Twilio Voice, Fastify, AWS — **Apr 2025**

- Engineered bidirectional voice streaming (<2s latency) with dynamic form-filling to streamline food assistance workflows; Runner-Up, UMD AI Case Competition.

03 AWARDS & GRANTS

$250,000	**Winner**, XFoundry Xperience Competition · Press ↗	DefenX
$25,000	**1st Place**, Pitch Dingman Competition	Renfort
$9,000	**Global Winner**, ICC & NIUM Next In 2.0 Global Hackathon (22k+ entries, 119 countries) · Press ↗	StumpEye
$5,000	**1st Place**, MIT Hacking Medicine BeyondHack Accelerator	Renfort
$4,500	**Best in Show**, Pitch Dingman Luby Idea Track	Renfort
$2,500	**3rd Place**, Spatial Intelligence Hackathon, UMD Startup Shell × Ironsite	VISTA
$1,500	**Runner-Up**, UMD Smith School AI Case Competition	FeedForward

04 TECHNICAL SKILLS

Languages	Python, C++, C#, TypeScript, JavaScript, C, SQL, R
Frameworks & Systems	PyTorch, .NET 8, WPF, Fastify, Bun, React, Node.js, TensorFlow, OpenCV, CUDA, ONNX
AI / LLM Infrastructure	Agentic Workflows, Real-time Voice, Continual Learning, RAG, Fine-Tuning, Prompt Engineering
Cloud & DevOps	Enterprise Integrations (ServiceNow), GCP, AWS, Docker, Win32 Automation, CI/CD, Git, Velopack

05 PUBLICATIONS

Octopus: A Latent Diffusion Model for Enhanced Text-Driven Manipulation in Image Synthesis — **IEEE CVMI 2024**

06 EDUCATION

University of Maryland, College Park · M.S. Applied Machine Learning — College Park, MD · **Aug 2024 – May 2026**

Amrita Vishwa Vidyapeetham · B.Tech. Computer Science — Coimbatore, India · **Oct 2020 – May 2024**